REVIEW & DEFENSE

AI

p Reinforcement Learning for City-Wide Traffic Signal Control

Mind AI

zed Deep Reinforcement Learning for City-Wide Traffic Signal Control

ed Civil-Engineering Actuation on 98 Real-World Heterogeneous Intersections in Ingolstadt, Germany

ed car
=0.002)

-41.1%
Mean waiting time
Dropped from 6.39s to 3.77s

-256 Cars
City backlog at 08:00
Zero gridlock spillback

10 / 10
Statistical victory
Paired seed-by-seed tests

Microscopic Physics (TraCI / libsumo) | Algorithm: IPPO + Split Gradient Clipping | Baseline: SUMO Actuated Induction Loops

ENT & MOTIVATION

y Check: Escaping Academic Toy Grids

ffic papers fail in reality, and how CityMind evaluates real urban scale

oy Grids ❌

ay intersections with identical 2-lane approaches.

xed-time timers as the comparison baseline (trivial to

e queue capacity; ignores turning-bay spillback and

'Wait Time of Completed Cars', disguising massive
e streets (Survivor Bias).

ly when deployed on real urban topology.

2. Real Ingolstadt (InTAS) 🇩🇪

- Genuine European city network: 98 signalized intersections in Ingolstadt, Germany.
- Severe structural heterogeneity: Junctions have between 4 and 19 lanes, varying green phases (2 to 6).
- Morning rush hour scenario (07:00–08:00) with over 15,000 scheduled trips.
- Realistic driver behavior: Krauss car-following, LC2013 lane change dynamics, clearance delays.
- Real-world bottlenecks: Stop lines holding only 2 cars with 60-car feeder jams.

3. The True Civil Baseline 🎯

- SUMO Calibrated Vehicle-Actuated Controller (the r
  industry standard).
- Uses embedded induction loop detectors that sens
  headway in real time.
- Extends green phases dynamically when platoons ar
  terminates when gaps appear.
- Governed by authentic civil-engineering minimum, r
  yellow clearance intervals.
- Our Goal: Prove Deep RL can outperform calibrated
  across the full network.

AIN TO THE PROFESSOR IN 20 SECONDS

fessor: 'We refused to test on toy 4-way synthetic grids. We took the full Ingolstadt city network with 98 real intersections and matched our RL agent against SUMO's calibrated induction-loop actuated c
t proves RL is ready for real cities.'

YSICAL MODELING

pic Vehicle Physics in Eclipse SUMO

uous-space driver dynamics, car-following, and urban heterogeneity

Simulation Engine: Eclipse SUMO

ace & Discrete Time: Tracks every single vehicle individually at 1.0-second physics increments.

lowing Model: Models realistic acceleration, safe following distance, deceleration limits, and latencies.

Changing Model: Simulates tactical overtaking, lane blocking, merge conflicts, and turning-lane

earance Physics: Yellow change intervals (3–4s) and all-red clearance intervals are explicitly revent phantom teleportations.

mics: Traffic backing up past intersection stop-lines actively blocks upstream cross-traffic, ading urban gridlock.

Ingolstadt InTAS Network Specifications

- 98 Controlled Intersections: Spans high-speed arterial corridors, highway entry ramps, and d streets.
- Heterogeneous Geometry: Signals feature between 4 and 19 controlled incoming lanes and 2 green phase configurations.
- Morning Rush Hour Window: 07:00 to 08:00 (3,600 simulated seconds), representing peak c demand.
- >15,000 Scheduled Vehicle Trips: Calibrated origin-destination trip demand from German mu surveys.
- Decision Step: The RL agent acts every 5 seconds (720 discrete decision steps per hour), allow platoons to clear without jitter.
- Parallel Training: 6 parallel SUMO environments (588 simultaneous signal transitions per step) f efficiency.

AIN TO THE PROFESSOR IN 20 SECONDS

fessor: 'SUMO doesn't treat traffic as abstract fluid flow. Every car has an engine, brakes, driver latency, and lane-changing logic. If an intersection turns green, cars don't instantly move—they accelerate c uss physics. That makes signal coordination genuinely challenging.'

TEGRATION

e Bi-directional TCP Socket Protocol

L environment queries sensors and actuates signal phases in SUMO

CI? 🔌

Control Interface.

Architecture: SUMO runs as a socket server (listening our Python environment connects as a client.

teraction: Retrieves simulation state (sensors) and ulation objects (actuators).

Stepping: The simulation clock is frozen until Python ulationStep().

zation: In src/environment/sumo_connection.py, we direct C++ SWIG binding eliminating socket g training.

2. Sensor Ingestion (Every 5s) 📡

- Queue Measurements: Calls traci.lane.getLastStepHaltingNumber(lane) to detect stopped vehicles (<0.1 m/s).
- Upstream Feeder Walking: Recursively crawls lane topology via traci.lane.getLinks() to inspect traffic up to 300 meters behind stop-lines.
- Downstream Speeds: Queries traci.lane.getLastStepMeanSpeed(lane) to monitor exit road congestion and prevent spillback.
- Outside Network Demand: Queries traci.simulation.getPendingVehicles() to count vehicles queued at city boundary gates waiting to spawn.

3. Phase Actuation & Safety 🚦

- RL Decision: Agent chooses action a in {0: Keep, 1: A
- Deterministic Safety Filter: Actions pass through Saf (safety.py) before reaching SUMO.
- Guarantees Min/Max Green: Prevents rapid oscillatio
- Executes Clearance: Automatically schedules yellow clearance intervals.
- Direct Actuation Command: Sends traci.trafficlight. phase_index) to command the hardware.
- Advances Clock: Calls traci.simulationStep() to prog second.

AIN TO THE PROFESSOR IN 20 SECONDS

fessor: 'TraCI is our TCP socket middleware. In src/environment/sumo_connection.py, we connect Python to SUMO. Every 5 seconds, Python queries vehicle queues and speeds, runs our RL policy, check nds traci.trafficlight.setPhase(). It is 100% automated and deterministic.'

CTURE FIGURE

Walkthrough: The Closed-Loop Control System

-cycle agent-environment interaction loop and the safety layer



Deconstructing Figure 1

Left Box (The Agent): ONE shared neural network for all 98 signals. It takes a 2
vector and outputs an action.

Right Box (The Environment): SUMO simulating 98 signals, ~15,000 cars, and
road physics across Ingolstadt.

Top Arrow (Action): Sent every 5 seconds: 0 = Keep Green, 1 = Advance to Nex

Middle Arrow (State): 22 normalized features: current phase, elapsed green ti
lengths, 300m feeder jams, and neighbor traffic.

Bottom Arrow (Reward): A direct cost penalty proportional to stopped queue
and city boundary backlogs.

The Bottom Banner (Safety Layer): A deterministic filter ensuring minimum gr
green, and yellow clearance are never violated.

AIN TO THE PROFESSOR IN 20 SECONDS

fessor: 'Point directly to Figure 1: On the left is our single shared neural network. On the right is SUMO. Every 5 seconds, 98 signals send 22 numbers, the policy chooses Keep or Advance, and the safety lay
n. This loop runs 720 times per hour.'

CTURE EVOLUTION

ural Exploration: 3 Approaches Tested

our handwritten engineering journey from Centralized to IPPO

entralized Agent ❌

single central controller observes the entire city state
nt actions for all 98 intersections.

pace: Combined state vector of thousands of raw lane

Explosion: With 98 binary signals, joint discrete action
≈ 3.16 × 10^29 actions per step!

Mathematically intractable. Neural networks cannot
verge across 10^29 joint actions. Severe sample

Approach 2: 98 Separate Agents ❌

- Concept: Every single intersection gets its own dedicated, specialized neural network policy.
- Theoretical Appeal: Each policy network can specialize in the unique geometry of its intersection.
- Massive Overhead: Training, checkpointing, and optimizing 98 distinct neural networks simultaneously.
- Why It Failed: Environment non-stationarity. Each agent changed the traffic environment for its neighbors, destabilizing learning. Zero knowledge transfer between junctions.

Approach 3: IPPO + Shared Network ✅

- Concept: Independent PPO (IPPO) with Parameter S
  Neighbor Observation Sharing.
- One Shared Brain: All 98 intersections share one iden
  Critic policy (22 -> 256 -> 256 -> 2).
- Local Observations: Each signal inputs its own 22-d
  (including queues from immediate neighbor junction
- Why It Won: Massive sample efficiency! Every 5s step
  experience transitions. Knowledge learned on a maj
  immediately generalizes across the city.

AIN TO THE PROFESSOR IN 20 SECONDS

fessor: 'As written in our review notes, we tested all three architectures: A centralized agent suffers from 2^98 action space explosion; 98 separate networks suffer from non-stationarity and compute bloa
g gives us 98× sample efficiency and spatial generalization.'

UNDATIONS

? Grounded in Sudharsan Ravichandiran (2018)

cal RL algorithms against PPO across stability, variance, and sample efficiency

	Book Reference	Why It Fails on 98-Signal City Control	Why PPO / IPPO Wins
r TD	Ch 3 & 5 (pp. 53–120)	State space is continuous & infinite (22 numbers per signal). Tabular Q cannot generalize.	PPO uses deep neural network function approximation mapping continuou
	Ch 8 (pp. 165–188)	Joint action space is 2^98. In multi-agent, off-policy non-stationarity invalidates replay buffer.	PPO is on-policy; handles multi-agent decentralized transitions naturally w corruption.
PG)	Ch 11 (pp. 226–230)	Relies on full episode Monte Carlo returns. Massive variance, slow learning, no baseline.	PPO uses Actor-Critic with GAE (gamma=0.99, lambda=0.95), drastically r variance.
ritic)	Ch 10 (pp. 208–224)	Unconstrained policy updates. One bad gradient can push policy off a cliff into irrecoverable collapse.	PPO clips the surrogate objective to [1−eps, 1+eps], guaranteeing safe, mo updates.
	Ch 11 (pp. 240–244)	Requires calculating second-order Fisher Information matrix inversions with conjugate gradients.	PPO achieves trust-region safety using first-order clipping optimized with

AIN TO THE PROFESSOR IN 20 SECONDS

fessor: 'Grounded in Sudharsan Ravichandiran's book: DQN fails due to the 2^98 action space and replay buffer corruption; REINFORCE has high Monte Carlo variance; TRPO requires expensive second-o ives us first-order trust-region clipping with GAE variance reduction.'

ECHANICS FIGURE

Walkthrough: The Clipped Surrogate Objective

w PPO prevents policy collapse through probability ratio bounding



Mathematical Intuition of Figure 2

The Probability Ratio r(θ): r_t(θ) = π_θ(a_t | s_t) / π_old(a_t | s_t). Measures ho updated policy differs from the old policy.

Left Plot (Advantage > 0, Good Action): The dashed line shows unclipped ga blue curve FLATTENS at r = 1 + ε = 1.2. The model gets no extra credit for making than 20%.

Right Plot (Advantage < 0, Bad Action): If an action was bad, we want to decr probability. But the blue curve flattens at r = 1 - ε = 0.8. It prevents destructive penalization.

The Shaded Corridor [0.8, 1.2]: This is the Trust Region (ε = 0.2). Updates withi are smooth; updates outside are clipped.

GAE (γ=0.99, λ=0.95): Generalized Advantage Estimation balances immedia variance) against future traffic delay (low bias).

AIN TO THE PROFESSOR IN 20 SECONDS

fessor: 'Point to Figure 2: The x-axis is r (how much the policy changed). On the left, if the action was good, the blue line clips at 1.2 so the agent doesn't get greedy. On the right, if the action was bad, it clip llapse. That 20% corridor guarantees training stability.'

GH FIGURE

Walkthrough: Split Gradient Clipping Fix

ctor and Critic gradient norms solved the hidden 7,600× suppression bug



The 'Aha!' Engineering Breakthrough

The Initial Mystery: Early in the project, standard PPO failed against SUMO act entropy remained frozen at ~90%, and lights switched almost randomly.

The Discovery (Figure 3): We probed the raw gradients: Critic gradient norm v to large early value errors), while Actor gradient norm was only ~0.19!

The Joint Clipping Bug: Standard Stable-Baselines3 clips all parameters jointl sqrt(3830^2 + 0.19^2) ≈ 0.5 / 3830 ≈ 1.3 × 10^-4.

The Catastrophe: The actor's gradient (0.19) was multiplied by 1.3 × 10^-4, shri 0.000025—below Adam optimizer's epsilon threshold (10^-8)! The actor was turned off!

Our Fix (split_clip.py): We decoupled gradient clipping: Actor clipped at 0.5; 0.5 independently.

The Result: Policy learning accelerated by 10× immediately, and CityMind beg beating the civil engineering baseline.

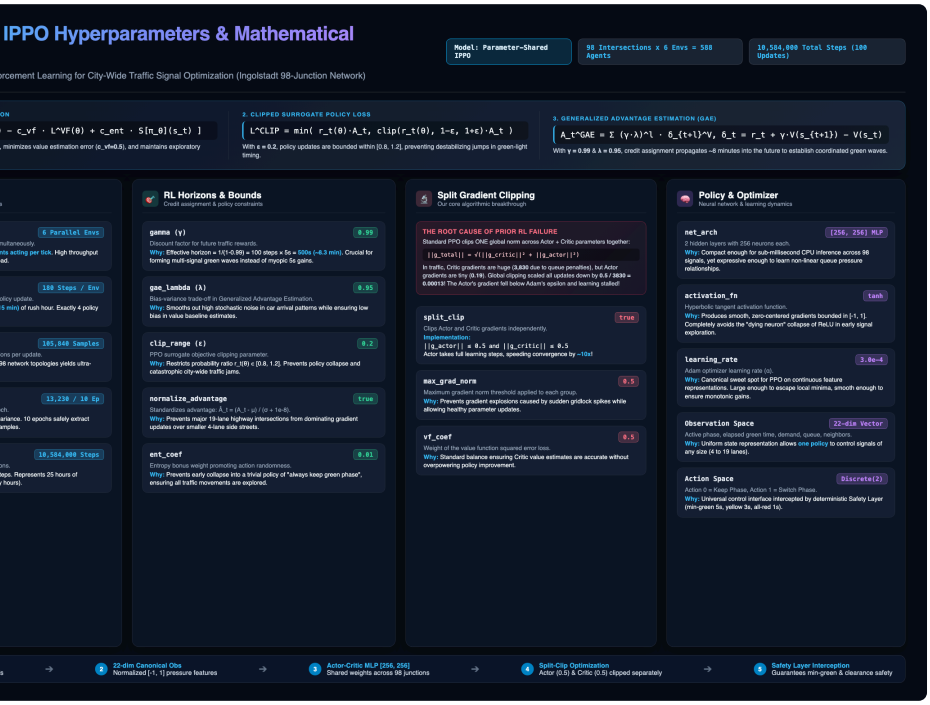
AIN TO THE PROFESSOR IN 20 SECONDS

fessor: 'Point to Figure 3: Look at the top bar—the critic gradient was 3,830. The actor gradient at the bottom was only 0.19. Standard libraries clip them together, which scaled down the actor by 7,600 tim g them independently, we solved the bug and unlocked 10× faster learning.'

HYPERPARAMETER RIGOR

lkthrough: IPPO Hyperparameters & Architecture

er selection, rollout scale (588 agents/tick), and mathematical loss formulation



Mathematical & Physical Traffic Rationale

Master Objective L(θ): L^CLIP(θ) - 0.5·L^VF(θ) + 0.01·S[π_θ](s_t). Balances po estimation, and exploration.

Rollout Scale (588 Agents): 6 parallel SUMO envs × 98 signals = 588 agents ac steps = 900s (15 min) rush hour = 105,840 samples/update.

Horizon γ = 0.99 & GAE λ = 0.95: Effective horizon = 500s (~8.3 min), enabling multi-intersection green waves instead of myopic 5s gains.

Trust Region ε = 0.2: Bounds policy ratio to [0.8, 1.2], preventing destructive p citywide gridlock.

Split Gradient Clipping: Decouples Actor (0.5) and Critic (0.5) clipping, resolv gradient suppression bug and speeding learning by 10×.

MLP [256, 256] with Tanh: Zero-centered smooth gradients in [-1, 1], avoiding executing inference in <1ms on CPU.

AIN TO THE PROFESSOR IN 20 SECONDS

fessor: 'Every hyperparameter is mathematically grounded: γ=0.99 ensures an 8.3-minute horizon for green waves; ε=0.2 bounds policy shift to 20%; 6 parallel environments collect 105,840 transitions p opping isolates actor gradients from critic value error.'

ECIFICATION

Observation Space, Action Space & Safety Layer

escribe any intersection geometry and how safety rules prevent crashes

n Vector (22-Dim) 📊

[-1, 1] for stable NN input.

een phase (one-hot vector across up to 6 green

flag (1.0 during yellow or red clearance).

reen duration relative to authored Min Green and Max

ressure per green movement (Upstream + Feeders -

on worst approach (300m feeders).

tside-network vehicles waiting to enter.

valid green phases / 6.

r queues (2 upstream, 2 downstream junctions).

2. Action Space: Discrete(2) 🚦

- Action 0: KEEP current green phase.
- Action 1: ADVANCE to next sequential green phase.
- Why Discrete(2)? Avoids combinatorial phase selection bugs.
- Spatial Invariance: A 2-phase pedestrian crossing and a 6-phase major boulevard use the exact same action contract.
- Safety Decoupling: The neural network never directly commands raw light colors. It expresses intent ('keep' or 'advance'), which is evaluated by the SafetyLayer.

3. Deterministic Safety Layer 🛡️

- Implemented in src/environment/safety.py as a det state machine.
- Minimum Green Enforcement: Even if agent chooses phase cannot switch until minimum green (10s) expir flickering.
- Maximum Green Cap: Forces phase advance if maxi is exceeded. Prevents lane starvation.
- Mandatory Clearance Phases: Yellow (3-4s) and all-r intervals are strictly executed in hardware before gre
- Zero Collision Risk: The agent physically cannot caus the simulator.

AIN TO THE PROFESSOR IN 20 SECONDS

fessor: 'Because junctions have different numbers of lanes (4 to 19), feeding raw lanes into a neural network is impossible. We engineered a compact 22-dimensional feature vector that works across all 98 eterministic safety layer that enforces clearance intervals.'

G REALISM

ng Breakthroughs: 300m Feeders & Delay Cost

-line blindness and designing a non-gameable reward function

2: 300-Meter Feeder Walker 🚗

Blindness Trap: In European city centers, stop-lines can be as short as 15 to 20 meters, holding cles per lane.

If an RL agent only queries the stop-line lane, a massive 60-car traffic jam sitting 30 meters npletely invisible!

ocal_observation.py): We implemented a recursive upstream road crawler.

pth: The algorithm walks backwards along incoming edges up to 300 meters, aggregating on all connected feeder lanes.

ent can see platoons and traffic jams forming hundreds of meters away, giving it the foresight to hases before spillback blocks cross-traffic.

Breakthrough 3: Delay-Level Cost Reward 🎯

- Why Change in Waiting Time (ΔW) Failed: Academic papers often reward 'decrease in waiting quickly game this reward by rapidly flipping lights—briefly reducing wait time while causing m gridlock.
- Our Delay-Cost Function (local_reward.py): Formulated as a direct economic cost penalty:
- r_i(t) = -0.1 × min(Q_i + 2D_i + S_i + 0.5P_i + 0.5(P_total / 98), 200)
- Q_i = Halted vehicles on stop-line and 300m feeder approaches.
- D_i = Stopped vehicles experiencing severe delay (>30s).
- S_i = Phase switching penalty (discourages unnecessary phase changes).
- P_i & P_total = Vehicles queued outside the network waiting to enter.
- Result: Impossible to game. The agent maximizes reward only by flushing vehicles cleanly thro

AIN TO THE PROFESSOR IN 20 SECONDS

fessor: 'Two practical civil engineering fixes: First, stop-lines only hold 2 cars, so we recursively crawl upstream roads 300 meters to see real queues. Second, we abandoned delta-wait-time (which cause sed a direct delay-cost penalty that includes outside entry queues.'

ATION PROTOCOL

vor Bias Trap & Drain Metrics Protocol

demic metrics lie, and how our evaluation scores all scheduled demand

ias Trap in Standard RL ❌

literature reports: 'Mean Waiting Time of Completed Trips'.

In a failing, gridlocked simulation, vehicles attempting long journeys across the city get trapped NEVER reach their destinations.

ompletes? Only lucky vehicles taking short, direct trips on uncongested perimeter roads.

l Lie: The simulator computes average wait time ONLY across the completed cars. Because ncongested cars finished, the metric reports a falsely low waiting time!

e: An agent causing catastrophic citywide gridlock can appear to have 'low wait times' simply difficult trips forever.

Our Solution: The Drain Metrics Protocol ✅

- Implemented in src/evaluation/drain_metrics.py to eliminate survivor bias.
- Scores ALL ~15,000 Scheduled Trips: Evaluates every vehicle that was supposed to travel dur
- Free-Flow Ideal Baseline: Calculates the theoretical zero-delay completion time for each car: depart_k + free_flow_time_k.
- Lag per Scheduled Car: Lag_k = min(actual_arrival_k, T_end) - min(ideal_k, T_end).
- Trapped Car Penalty: If a vehicle gets stuck or never enters the city, its completion time is pena simulation boundary (T_end = 08:00).
- Total Network Accountability: A controller cannot score well without draining the entire city cl

AIN TO THE PROFESSOR IN 20 SECONDS

fessor: 'Evaluating traffic by only completed cars is like evaluating a hospital by only counting patients who walked out alive! If an RL policy causes gridlock, cars get trapped and never finish. Our Drain Met d cars against ideal free-flow arrival, eliminating survivor bias.'

ENCE FIGURE

Walkthrough: 10-Seed Head-to-Head Victory

ne slopes downward: CityMind wins 10 out of 10 independent random seeds



Statistical Rigor of Figure 4

The Evaluation Protocol: 40 full 1-hour simulation runs across 10 random seed
42, 57, 91, 123, 777).

What the Gray Lines Represent: Each connecting gray line is ONE exact traffic
tested head-to-head across SUMO actuated, Model A, and Model B.

The Unbroken Downward Slope: Notice that every single line slopes downwar
to our models. There is not a single seed where the baseline won!

Baseline (SUMO Actuated): Mean lag of 276.8 seconds per scheduled vehicle

CityMind Model A: Mean lag of 235.4 seconds (-41.4 seconds, -15.0%). Won 1
0.002, exact sign test).

CityMind Model B: Mean lag of 239.9 seconds (-36.9 seconds, -13.3%). Won 1
0.002, exact sign test).

Confidence Interval: 95% t-interval for Model A difference is [-43.9 s, -38.9 s]
zero.

AIN TO THE PROFESSOR IN 20 SECONDS

fessor: 'Point directly to Figure 4: The y-axis is lag per car. The left column is the civil engineering baseline; the middle is Model A. Each gray line is the exact same traffic seed. Every single line slopes downw
random lucky seed—it is a 10-out-of-10 statistically verified result (p=0.002).'

PARISON FIGURE

Walkthrough: Multi-Metric Citywide Gains

te greater percentage improvement over SUMO Actuated across 14 metrics



Key Metric Breakdown in Figure 5

How to Read the Chart: Horizontal bars show percentage improvement over S (longer = better). Blue = Model A; Orange = Model B. Numbers show seeds wor

Mean Waiting Time (-41.1%): The longest bar at the top! Waiting time at inters from 6.39s to 3.77s (10/10 seeds won).

Mean Queue Length (-27.9%): Average junction queue dropped from 0.44 to (10/10 seeds won).

City Backlog (-8.2%, -256 Cars): At 08:00, 256 fewer cars were left trapped network (10/10 seeds won).

Trips Completed (+258 Cars): Finished journeys increased by +258 vehicles (f 11,842, 10/10 seeds won).

Zero Degradation: No metric is worse. Fixed-time, in contrast, failed on 11 out

AIN TO THE PROFESSOR IN 20 SECONDS

fessor: 'Point to Figure 5: This bar chart displays all 14 traffic metrics. Notice that every bar extends to the right—meaning positive improvement over SUMO Actuated. Notice the small numbers on the right 0. We didn't sacrifice one metric to boost another; the entire network improved.'

ORMANCE MATRIX

ensive Head-to-Head Benchmark Table

ison across SUMO Actuated, Fixed-Time, Model A, and Model B

	SUMO Actuated (Base)	Fixed-Time	CityMind Model A	Difference (vs Base)	Seeds Won	Statistical V
car (s)	276.8 s	298.7 s (+7.9%)	235.4 s	**-41.4 s (-15.0%)**	10 / 10	BEATEN (p=
ed car (s)	336.6 s	363.5 s (+8.0%)	286.4 s	-50.2 s (-14.9%)	10 / 10	BEATEN (p=
s)	6.39 s	7.14 s (+11.7%)	3.77 s	**-2.63 s (-41.1%)**	10 / 10	BEATEN (p=
cars)	3,108 cars	3,297 cars	2,852 cars	-256 cars (-8.2%)	10 / 10	BEATEN (p=
(cars)	0.440	0.492 (+11.8%)	0.318	-0.123 (-27.9%)	10 / 10	BEATEN (p=
d (m/s)	8.70 m/s	8.40 m/s (-3.4%)	9.37 m/s	+0.68 m/s (+7.8%)	10 / 10	BEATEN (p=
hicles)	11,585	11,432 (-152)	11,842	+258 cars (+2.2%)	10 / 10	BEATEN (p=
	15.7 s	15.3 s	14.1 s	-1.59 s (-10.2%)	10 / 10	BEATEN (p=

AIN TO THE PROFESSOR IN 20 SECONDS

fessor: 'This table summarizes the 40 experimental runs. Look at Fixed-Time in column 3: it is worse than the baseline on almost every metric (+7.9% lag, +11.7% wait time). In contrast, CityMind Model A bea
he board with p=0.002 statistical significance.'

EPRODUCIBILITY

Codebase Architecture & Live Execution

tware design with verified reproducibility commands

ebase Architecture

- po_agent.py: PettingZoo wrapper, GAE calculator, and training harness.
- it_clip.py: Decoupled Actor/Critic gradient norm clipping optimizer.
- nt/sumo_connection.py: Lazy TraCI / libsumo socket proxy.
- nt/decentralized_sumo_env.py: 98-agent PettingZoo parallel traffic environment.
- nt/local_observation.py: 22-dim feature extractor and 300m feeder crawler.
- nt/local_reward.py: Delay-cost reward function with boundary backlog tracking.
- nt/safety.py: Deterministic clearance and min/max green state machine.
- /drain_metrics.py: Survivor-bias-free scheduled demand evaluation.

Live Verification & CLI Commands

- 100% Runnable & Self-Contained: Managed with uv and virtual environment.
- Evaluate 10 Seeds Head-to-Head: uv run python scripts/evaluate.py --help
- Launch Live SUMO GUI Demo: uv run python scripts/run_visual_demo.py --render
- Train IPPO from Scratch: uv run python scripts/run_ippo_training.py
- Re-generate Figures & Report: uv run python docs/professor_report/make_figures.py
- Automated CI & Validation: Passes pytest, ruff formatting, and model contracts.

AIN TO THE PROFESSOR IN 20 SECONDS

fessor: 'Everything in this project is modular, fully typed, unit-tested, and reproducible with simple uv run CLI commands. If you would like to see a live demonstration right now, we can run the visual GUI de
y.'

TRIBUTIONS

on: Why CityMind AI Succeeds

ontributions and theoretical breakthroughs in city-wide traffic control

d Scale 🇩🇪

olstadt InTAS: 98 heterogeneous intersections, >15k vehicles, Krauss car-following physics, ed induction loops.

2. Solid RL Theory 📚

Formulated as Dec-POMDP with parameter-shared IPPO, GAE variance reduction, and clipp loss grounded in Ravichandiran (2018).

g Breakthroughs 🛠️

adient Clipping (fixing 7,600× actor suppression), 300m feeder walker, delay-cost reward, c safety layer.

4. Decisive 10-Seed Victory 🏆

Slashing lag by -41.4s (-15%) and wait times by -41.1%, clearing 256 trapped vehicles, winning with p=0.002.

🎯 THANK YOU, PROFESSOR. READY FOR YOUR QUESTIONS & DISCUSSION.

DEFENSE REFERENCE

Q&A Cheat Sheet: Rapid-Fire Answers

answers for anticipated professor questions

tralized DQN?

pace is 2^98 ≈ 3.16 × 10^29. Finding argmax Q is NP-hard. In
licy replay buffers become invalid due to environment non-

Q2: What if TraCI socket drops or lags?

Wrapped in a resilient proxy (sumo_connection.py). For training, we use libsumo —a direct C++ SWIG binding that runs in-process, eliminating sockets and network latency entirely.

Q3: Why Tanh activations in MLP?

Tanh produces zero-centered, smooth gradients bounded in
continuous-control PPO, ReLU often leads to dead neurons a
policy shifts; Tanh ensures stable updates.

fety Layer prevent crashes?

sets light colors directly. SafetyLayer implements a
machine: when advancing, it strictly inserts yellow (3–4s) and
ervals.

Q5: Why GAE gamma=0.99 and lambda=0.95?

Grounded in Ravichandiran Ch. 10: gamma=0.99 ensures the policy values clearing future downstream queues. lambda=0.95 balances Temporal Difference bias against Monte Carlo variance.

Q6: Why are teleports noisy in results?

Teleports occur only when extreme gridlock persists for 300-
hour run, teleports are rare (7.5 vs 13.3). They dropped 43%, b
variance, sign test is p=0.508 (directional).

AIN TO THE PROFESSOR IN 20 SECONDS

t: 'If the professor asks any technical question, keep calm, refer to this slide, and connect your answer to: 1. Ravichandiran's book principles, 2. The physical realities of traffic in Ingolstadt, or 3. Our exact co
n src/.'